Midterm Progress Report

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Soccer Analytics Capstone — Trilemma Foundation
Georgia Tech MSA Spring 2026 Practicum

Project Overview

Goal: Build an open-source analytics pipeline that ingests public match event data and produces interactive player/team analytics dashboards.

Data Sources:

- StatsBomb: 3,464 matches, 12.2M events across major European leagues (La Liga, Premier League, Serie A, Ligue 1, Bundesliga) and international competitions
- Polymarket: 8,549 betting markets with $3B+ total trading volume, 1.1M individual trades spanning April 2025 — January 2026

Key Challenge: IDs are NOT normalized across datasets — StatsBomb team IDs do not map to Polymarket market slugs. Entity resolution is a critical early task.

Track Selected: Track 2 — Soccer Analytics Dashboard

Work Completed: Infrastructure & Data Pipeline

Repository Setup

- Forked official template repo, installed all dependencies
- Downloaded and organized both StatsBomb and Polymarket datasets
- Verified data integrity with starter EDA template

Custom Analytics Modules (3 Python modules, ~1,400 LOC)

- entity_resolution.py — Fuzzy team name matching between StatsBomb and Polymarket (150+ teams mapped)
- feature_engineering.py — Possession chain builder, xG flow, PPDA, Field Tilt, Expected Threat (xT), team style classifier
- market_analysis.py — Market efficiency metrics, volume analysis, odds volatility, competition breakdown

Enhanced Dashboard

- Extended template with Top Players by xG and Most Active Players charts
- Updated footer and branding for 2026

Work Completed: EDA Key Findings

StatsBomb Event Analysis

- Pass completion rate: 77.7% overall — uniform across competitions
- 88,023 shots with xG data; mean xG per shot = 0.10
- Average 2.85 goals per match; La Liga shows highest scoring
- Messi leads with 363.4 cumulative xG from 2,670 shots

Polymarket Market Analysis

- Champions League Winner markets: >$1B volume (33% of total)
- Trading peaks during European evening hours (17:00–22:00 UTC)
- Buy/Sell ratio: 82% BUY vs 18% SELL — strong bullish sentiment
- Only 38% of markets have any trade activity

Work Completed: Predictive Modeling Baseline

Research Question: Does event-level shot quality predict match outcomes?

Model: OLS regression — goal_diff ~ xg_diff

- goal_diff = home_score − away_score
- xg_diff = home_xG − away_xG

Results (3,457 matches):

- Slope = 1.055 — each unit of xG advantage translates to ~1 goal advantage
- R^2 = 0.475 — xG differential explains 47.5% of variance in actual goal differential
- Intercept ≈ 0.045 — minimal home advantage effect in xG-adjusted model

Interpretation: A single feature derived entirely from event data explains nearly half of outcome variance. This validates xG-based features as a foundation for dashboard analytics.

Work Completed: Advanced Metrics Pipeline

Possession Chains — sequence of events by same team until turnover

Segments matches into possession sequences with chain-level summaries

PPDA (Passes Per Defensive Action) — measures pressing intensity

Lower PPDA = more intense pressing (High < 10, Medium 10–15, Low > 15)

Field Tilt — territorial dominance via final-third passing share

Percentage of total final-third passes belonging to each team

Expected Threat (xT) — positional value using 12×8 grid

Pre-computed threat values increase toward opponent's goal

Work Completed: Entity Resolution

Challenge: StatsBomb and Polymarket use different team identifiers

e.g., "Arsenal FC" (StatsBomb) vs "arsenal" (Polymarket slug)

Approach:

Name normalization — remove common suffixes (FC, CF, SC), normalize punctuation

Team extraction from Polymarket questions — regex pattern matching

Fuzzy matching — token overlap (Jaccard similarity) with configurable threshold

Manual overrides for known mappings (15 major clubs)

Results:

308 unique StatsBomb teams identified

1,863 unique Polymarket team mentions extracted

1,699 fuzzy match mappings generated

Coverage: major European clubs well-covered; smaller/non-European clubs limited

Plan: Remaining Work

1. Dashboard Development (Primary Deliverable)

    - Interactive Dash application with xG flow visualization for individual matches
    - Team style comparison views (radar charts, scatter plots)
    - Player performance analytics (heatmaps, pass networks, xG leaders)
    - Competition-level tactical metrics explorer (PPDA, Field Tilt across seasons)

2. Multi-Feature Predictive Modeling

    - Extend baseline OLS model with PPDA, field tilt, pressure events, carry distance
    - Forward feature selection to identify most informative event metrics
    - Test temporal stability — do coefficients hold across seasons/competitions?

3. Market Comparison Analysis (Optional/Stretch)

    - For matches with Polymarket coverage, compare model-implied probabilities vs. market odds
    - Identify potential market inefficiencies around key events (goals, red cards)

Timeline & Milestones

Completed (Weeks 1–6)

- Data pipeline setup and validation
- Entity resolution module
- Feature engineering pipeline (PPDA, xG flow, Field Tilt, xT, team styles)
- Market analysis module
- EDA deliverables (Executive + Technical notebooks)
- Baseline predictive model ($R^2 = 0.475$)

In Progress (Weeks 7–10)

- Interactive dashboard — core views and filtering
- Multi-feature modeling and evaluation
- Documentation and README updates

Planned (Weeks 11–14)

- Dashboard polish — responsive design, deployment
- Market comparison analysis (if time permits)
- Final report and presentation

Questions?

Repository: github.com/kmylavarapu3/soccer-analytics-capstone-template